Electronic Devices & Circuits Laboratory

Experiment 1

V–I Characteristics of PN Junction & Zener Diodes,
Clipper and Clamper Circuits

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1 Aim

- To plot the forward and reverse V–I characteristics of a PN junction diode and find its cut-in voltage, static and dynamic resistance, and reverse saturation current.
- To plot the forward and reverse V–I characteristics of a Zener diode and find its breakdown voltage V_Z and Zener resistance R_Z.
- To design and test positive, negative and biased clipper circuits, and positive and negative clamper circuits, and record the input-output waveforms.

2 Components Used

Component	**Value/Part**	**Qty**
PN junction diode	1N4007 / 1N4148	1
Zener diode	1N4733A ($V_Z = 5.1$ V)	1
Regulated DC supply	0–30 V	1
Function generator	0–1 MHz	1
CRO / DSO	20 MHz	1
Digital multimeter	–	2
Resistors	1 kΩ, 10 kΩ, 100 kΩ	as needed
Capacitor	10 μF	1
Breadboard, wires, probes	–	as needed

3 PN Junction Diode

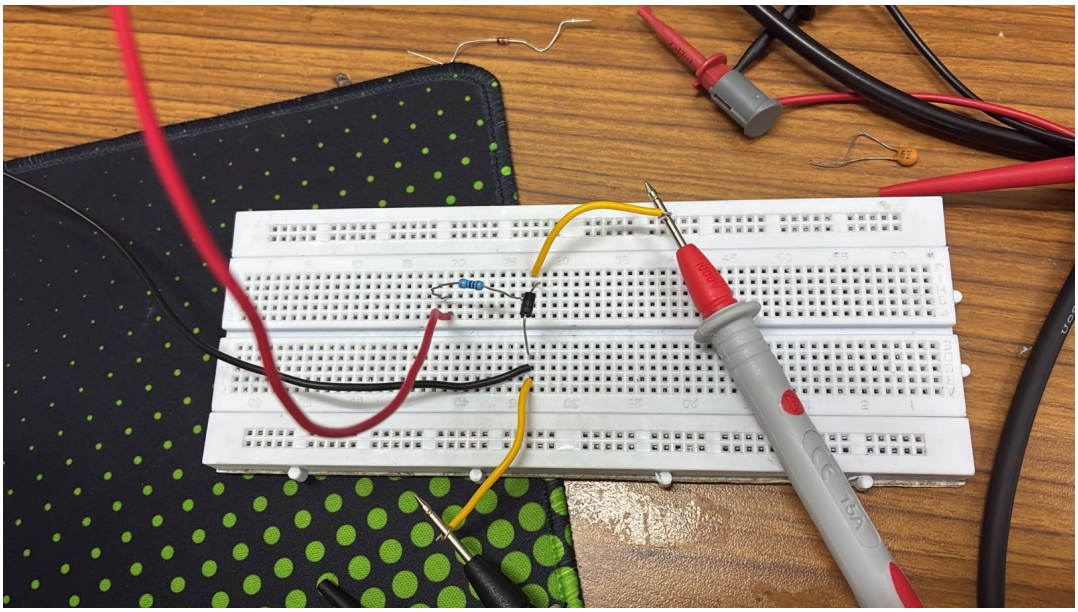


Figure 1: Forward Biased

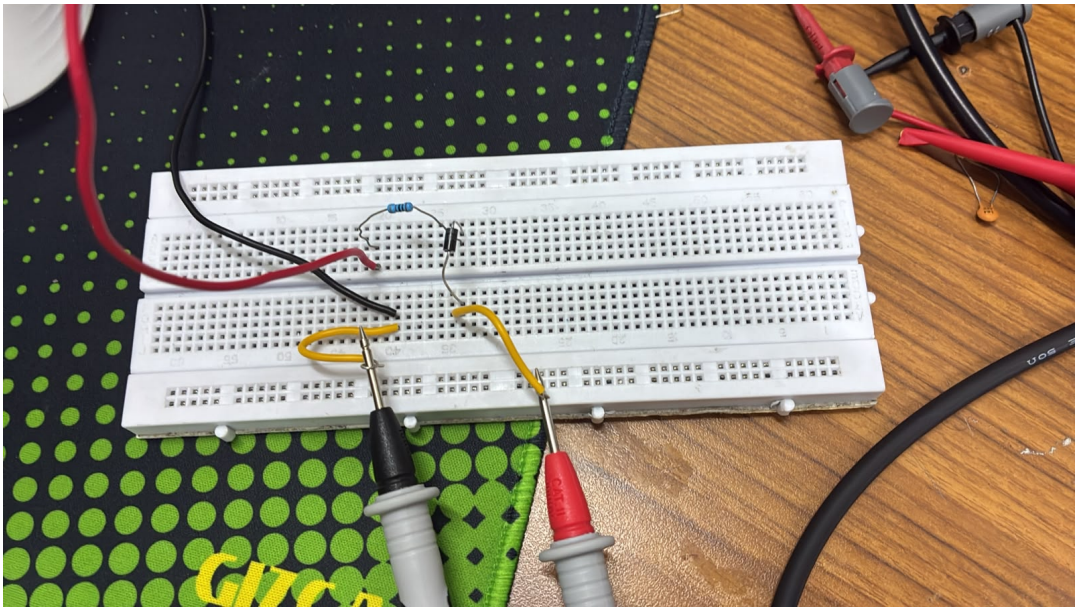


Figure 2: Reverse Biased

3.1 Theory

A PN junction diode conducts mainly in one direction. In forward bias, once the applied voltage crosses the cut-in voltage V_γ (about 0.7 V for silicon), the current starts rising very fast for a small increase in voltage, because majority carriers are being injected across the junction in large numbers. In reverse bias, only a small reverse saturation current I_0 flows due to minority carriers, and this stays almost constant until breakdown happens at a large negative voltage.

The diode current is approximately given by the diode equation:

$$I = I_0 \left(e^{V/\eta V_T} - 1 \right) \tag{1}$$

where V_T is the thermal voltage ($\approx$ 26 mV at room temperature) and η is the ideality factor.

From the forward characteristic we can find:

- **Static resistance** at a point: $R_{DC} = V/I$
- **Dynamic resistance** near a point: $R_{AC} = \Delta V/\Delta I$

3.2 Circuit and Simulation (Forward Bias)

The circuit was simulated in LTSpice with a 1 kΩ series resistor and the supply swept from 0 to 0.5 V in steps of 0.1 V, followed by finer steps near the knee.

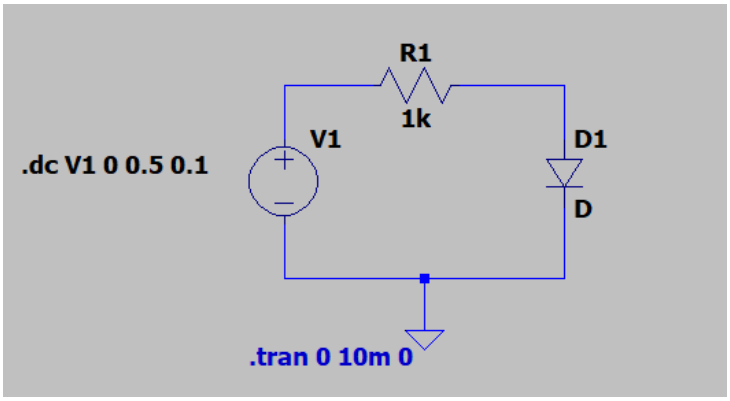


Figure 3: LTSpice circuit for PN junction diode forward bias test

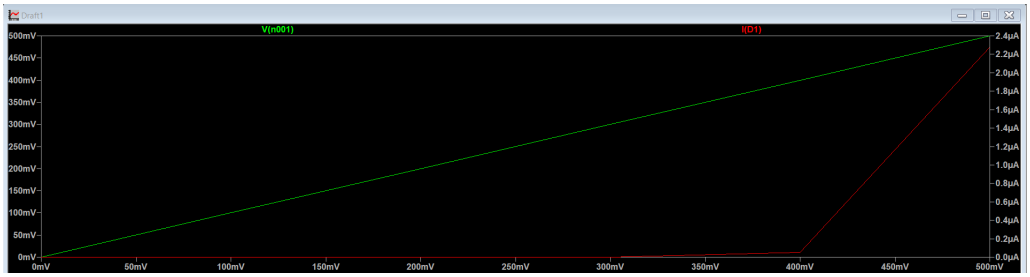


Figure 4: Simulated V–I plot: voltage across diode (green) and diode current (red) vs. sweep voltage. Current stays near zero till about 0.4 V and then rises sharply, showing the cut-in behaviour.

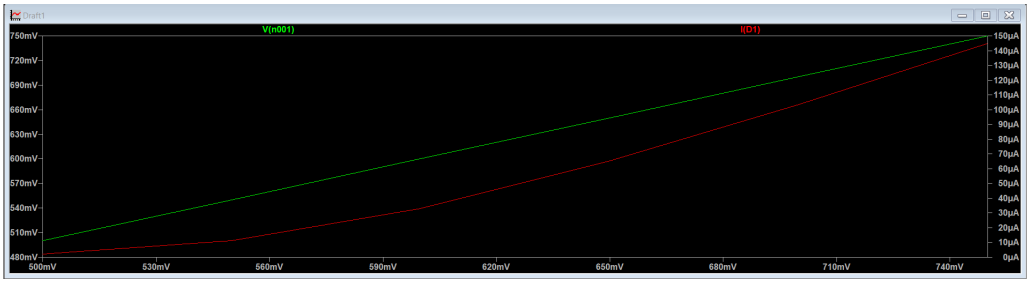


Figure 5: Simulated forward V–I characteristic, second view

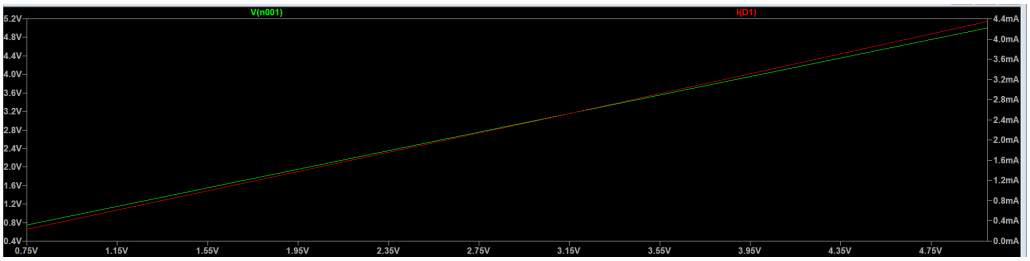


Figure 6: Simulated forward V–I characteristic, third view

From the simulation, the cut-in voltage is close to 0.4–0.5 V for the SPICE model used

(this will be verified against the measured value from the actual diode in lab, which is usually closer to 0.6–0.7 V for silicon).

3.3 Observation Table – Forward Bias

Table 1: Forward Bias Diode Characteristics and Power Source Voltage Measurements

Diode Voltage V_f (V)	**Diode Current I_f (mA)**	**Power Source Voltage (V)**
0	0	0
0.1	0	0.1
0.2	0.0001	0.2
0.3	0.0001	0.3
0.397	0.0053	0.4
0.472	0.0384	0.5
0.496	0.0674	0.55
0.5135	0.1017	0.6
0.5266	0.139	0.65
0.537	0.1793	0.7
0.5466	0.2207	0.75
0.635	2.02	2.64
0.663	4	4.64
0.678	5.995	6.65
0.689	8	8.64
0.698	9.9994	10.62
0.706	12.02	12.61
0.7109	14.05	14.61
0.716	16.009	16.5
0.72	18.01	18.43
0.723	20	20.33

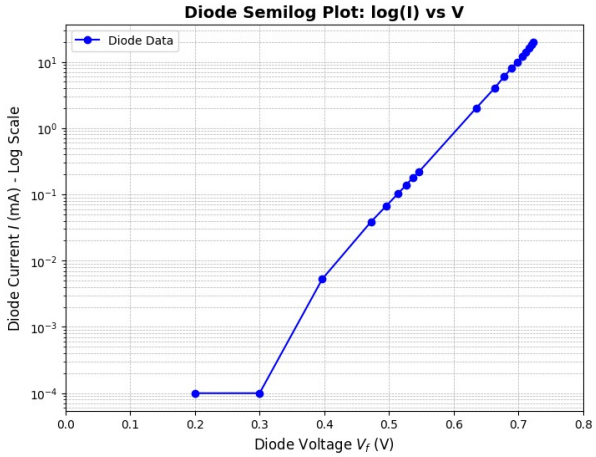


Figure 7: Semi-Log Graph

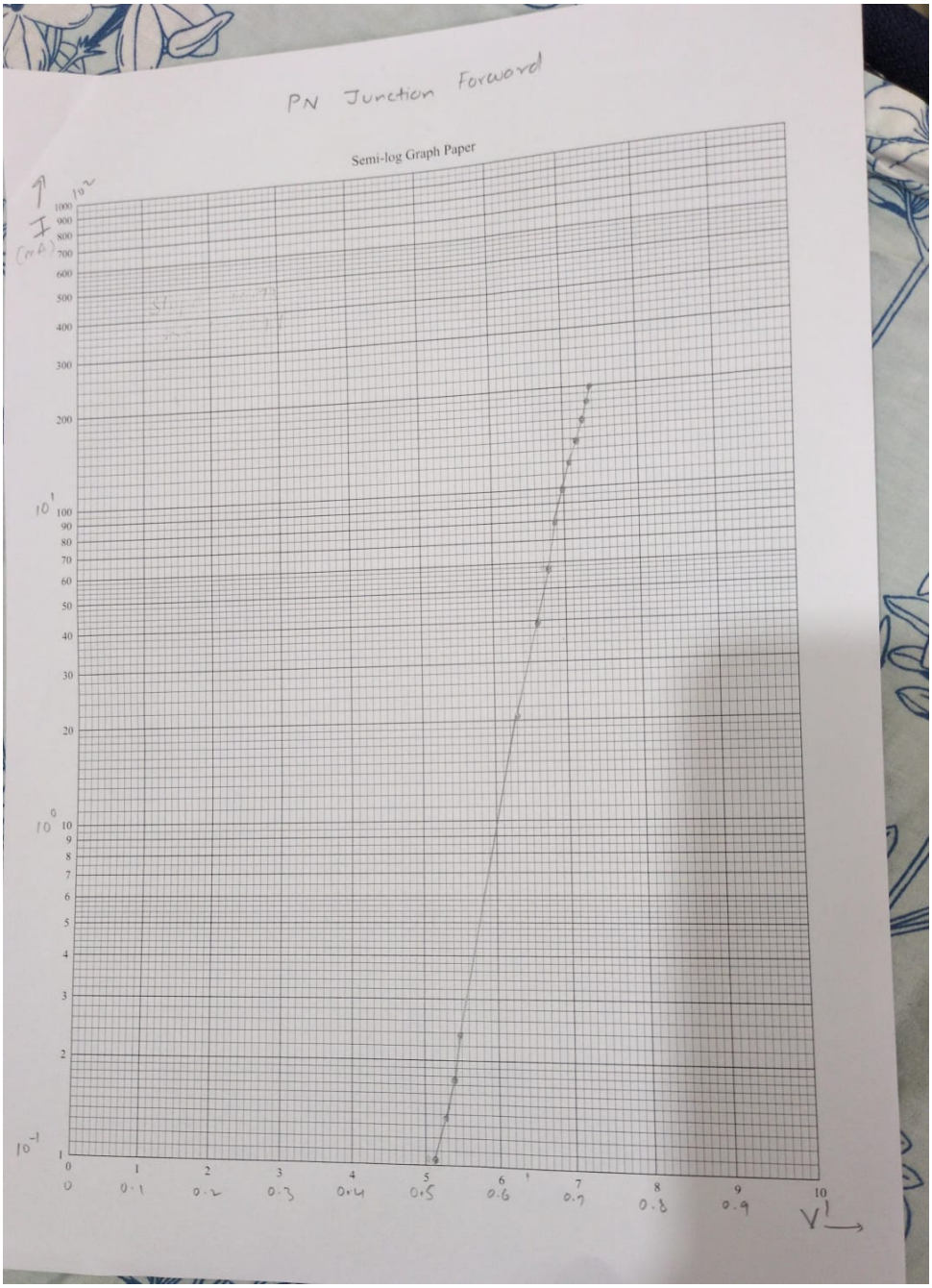


Figure 8: Semi-Log Graph

Static Resistance (at $V_f = 0.723$ V, $I_f = 20$ mA) $= \dfrac{V_f}{I_f} = 36.15\,\Omega$

Dynamic Resistance (near this operating point) $= \dfrac{\Delta V}{\Delta I} = 1.51\,\Omega$

From Table 1, the current rises sharply between $V_f = 0.397$ V and $V_f = 0.472$ V, giving an experimental cut-in voltage of

$$V_\gamma \approx 0.5 \text{ V}$$

This is somewhat lower than the typical silicon value of 0.7 V, likely due to the specific 1N4007/1N4148 diode model and measurement resolution near the knee.

3.4 Observation Table – Reverse Bias

No simulation was done for this part since the reverse saturation current of a real diode model is very small and not reliably captured in LTSpice's default diode model. This will be measured directly in the lab using a microammeter.

Table 2: Reverse Bias Diode Characteristics and Power Source Voltage Measurements

Diode Voltage Vf (V)	**Diode Current (mA)**	**Power Source Voltage (V)**
0	0	0
4.99	0	5
9.97	0	10
14.964	0	15
19.95	0	20
24.931	0.0001	25
29.919	0.0001	30

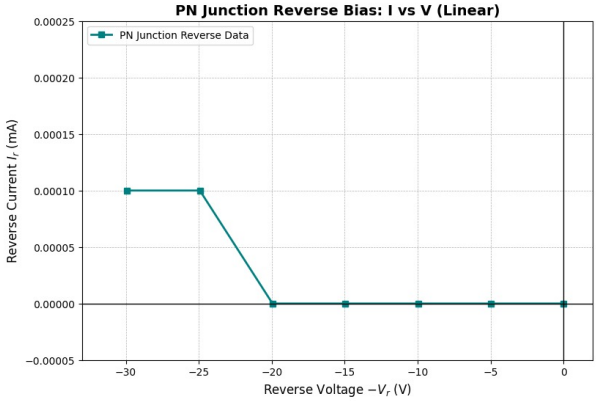


Figure 9: Linear Graph

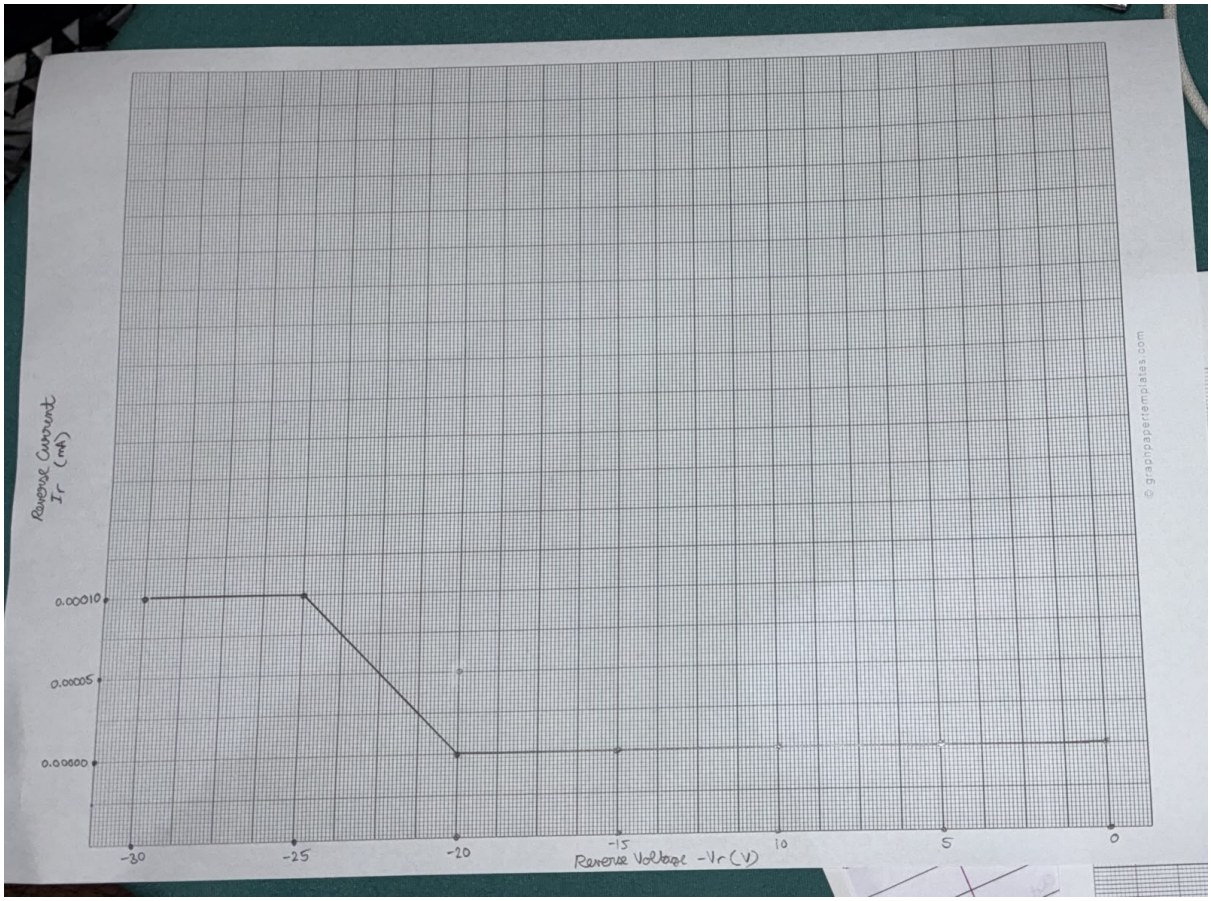


Figure 10: pn junction diode reverse current vs voltage

4 Zener Diode

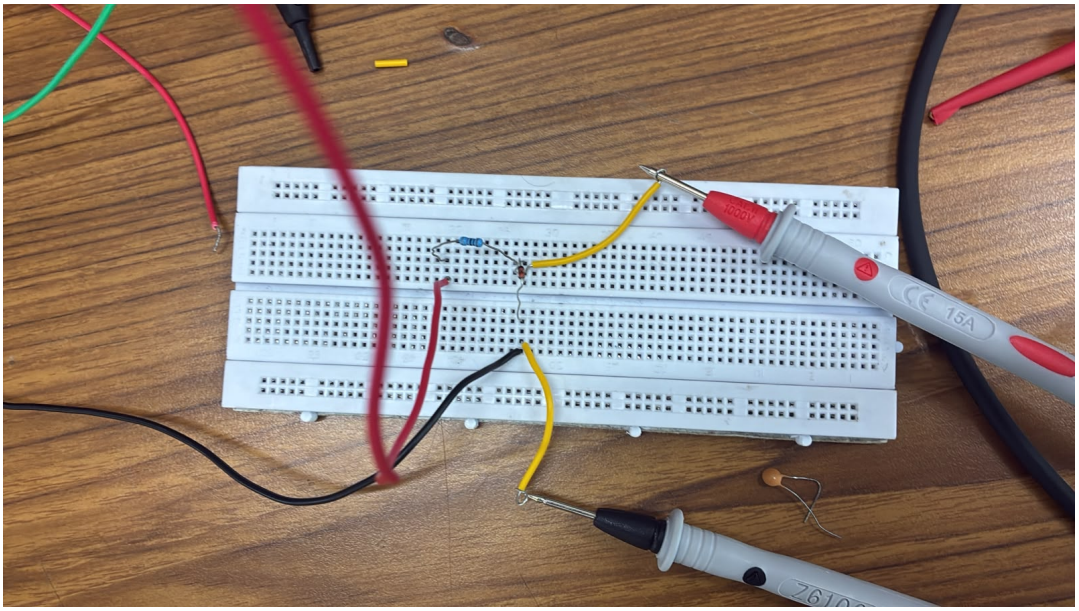


Figure 11: Positive Bias Zener Circuit

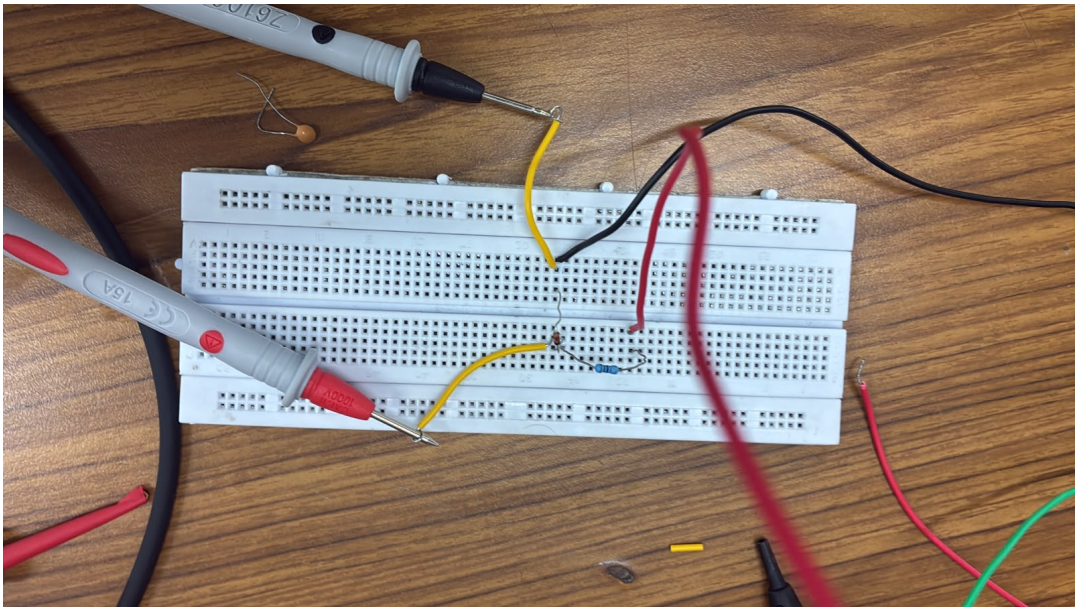


Figure 12: Negative Bias Zener Circuit

4.1 Theory

A Zener diode is a heavily doped PN junction meant to be operated in reverse breakdown without getting damaged. Its forward characteristic is the same as a normal diode. In reverse bias, once the voltage reaches the Zener voltage V_Z, the current increases sharply while the voltage stays almost constant – this near-vertical knee is what makes it useful as a voltage reference. Breakdown below about 5 V is mainly due to the Zener effect (field emission), and above about 6 V it is mainly avalanche multiplication.

The Zener resistance is calculated from the slope of the breakdown region:

$$R_Z = \frac{\Delta V_Z}{\Delta I_Z} \tag{2}$$

4.2 Circuit and Simulation - Forward Bias

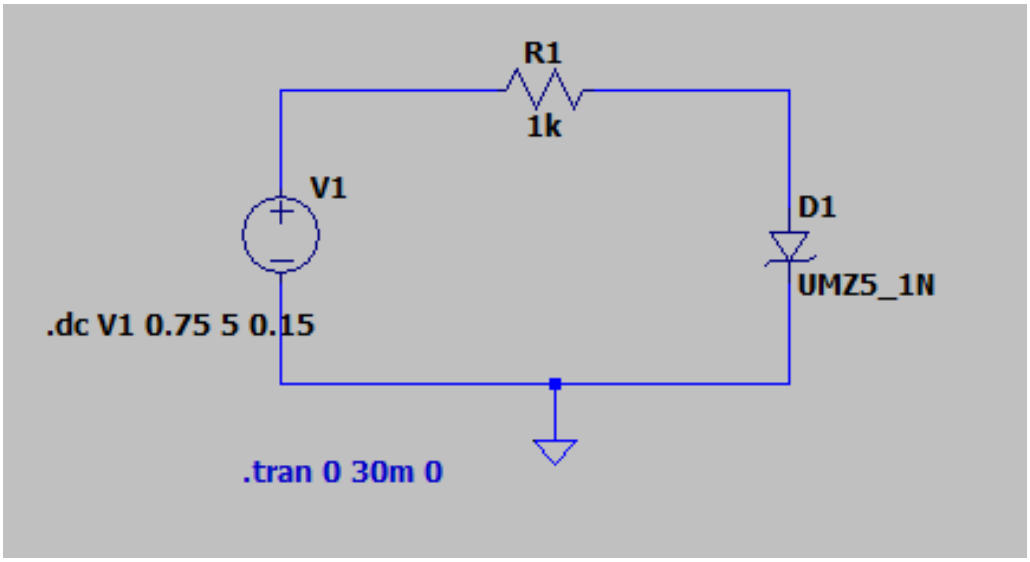


Figure 13: Forward Biased Zener Diode Circuit

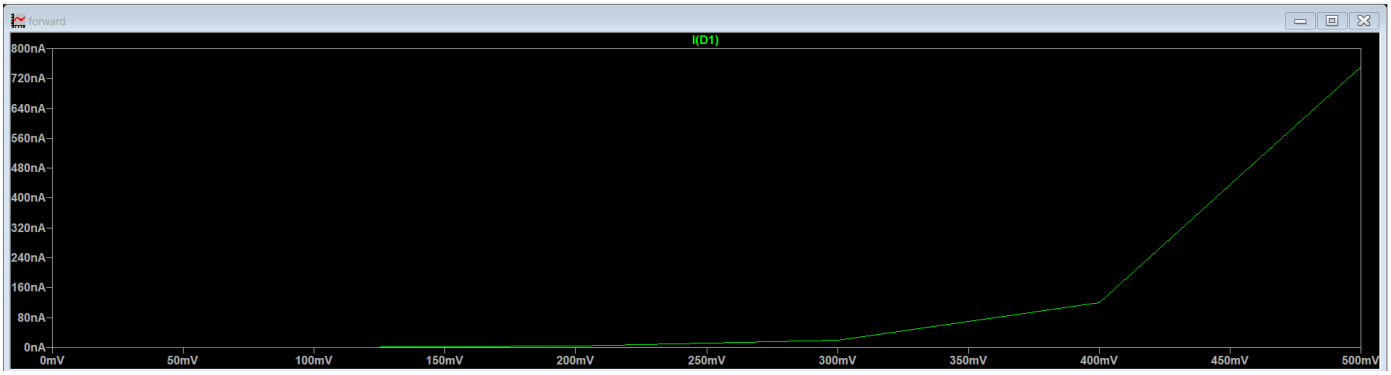


Figure 14: Simulation 1 - LTSpice

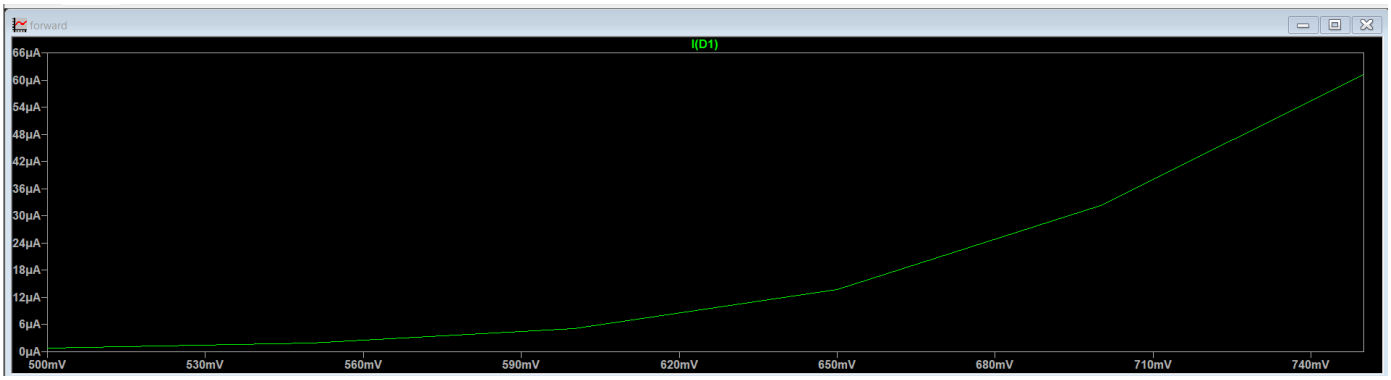


Figure 15: Simulation 2 - LTSpice

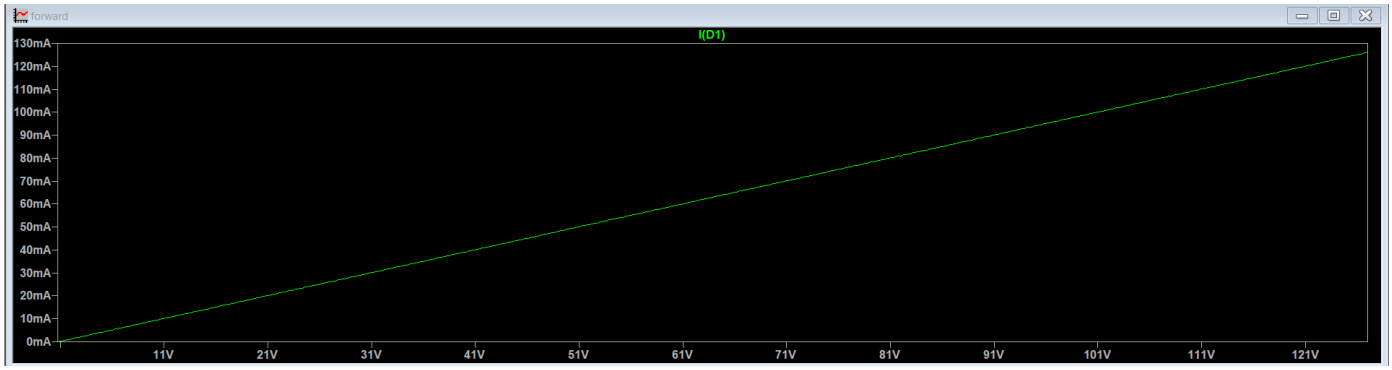


Figure 16: Simulation 3 - LTSpice

Table 3: Zener Diode Forward Bias Characteristics and Source Voltage Measurements

Diode Voltage Vf (V)	**Diode Current (mA)**	**Power Source Voltage (V)**
0	0	0
0.1	0	0.1
0.2	0	0.2
0.3	0	0.3
0.4	0	0.4
0.5	0.0002	0.5
0.548	0.0016	0.55
0.592	0.008	0.6
0.625	0.025	0.65
0.645	0.055	0.7
0.66	0.0915	0.75
0.669	0.1315	0.8
0.676	0.174	0.85
0.717	0.785	1.5
0.73	1.275	2
0.743	2	2.74
0.7604	4.006	4.75
0.7706	5.998	6.74
0.779	8.005	8.74
0.785	9.995	10.7
0.7907	12	12.68
0.794	14.004	14.64
0.797	16.01	16.59
0.801	18	18.53
0.804	20	20.43

From Table 3, the forward current rises appreciably between $V_f = 0.592$ V and $V_f = 0.66$ V, giving a forward cut-in voltage of

$$V_\gamma \approx 0.6 \text{ V}$$

This is slightly lower than the typical silicon PN junction value of 0.7 V, consistent with the heavier doping used in Zener diodes.

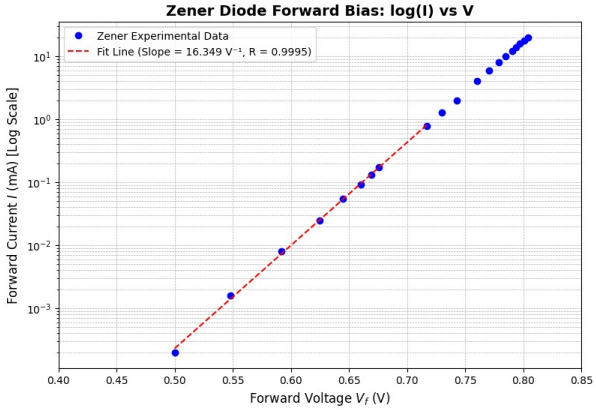


Figure 17: Semi-Log Graph

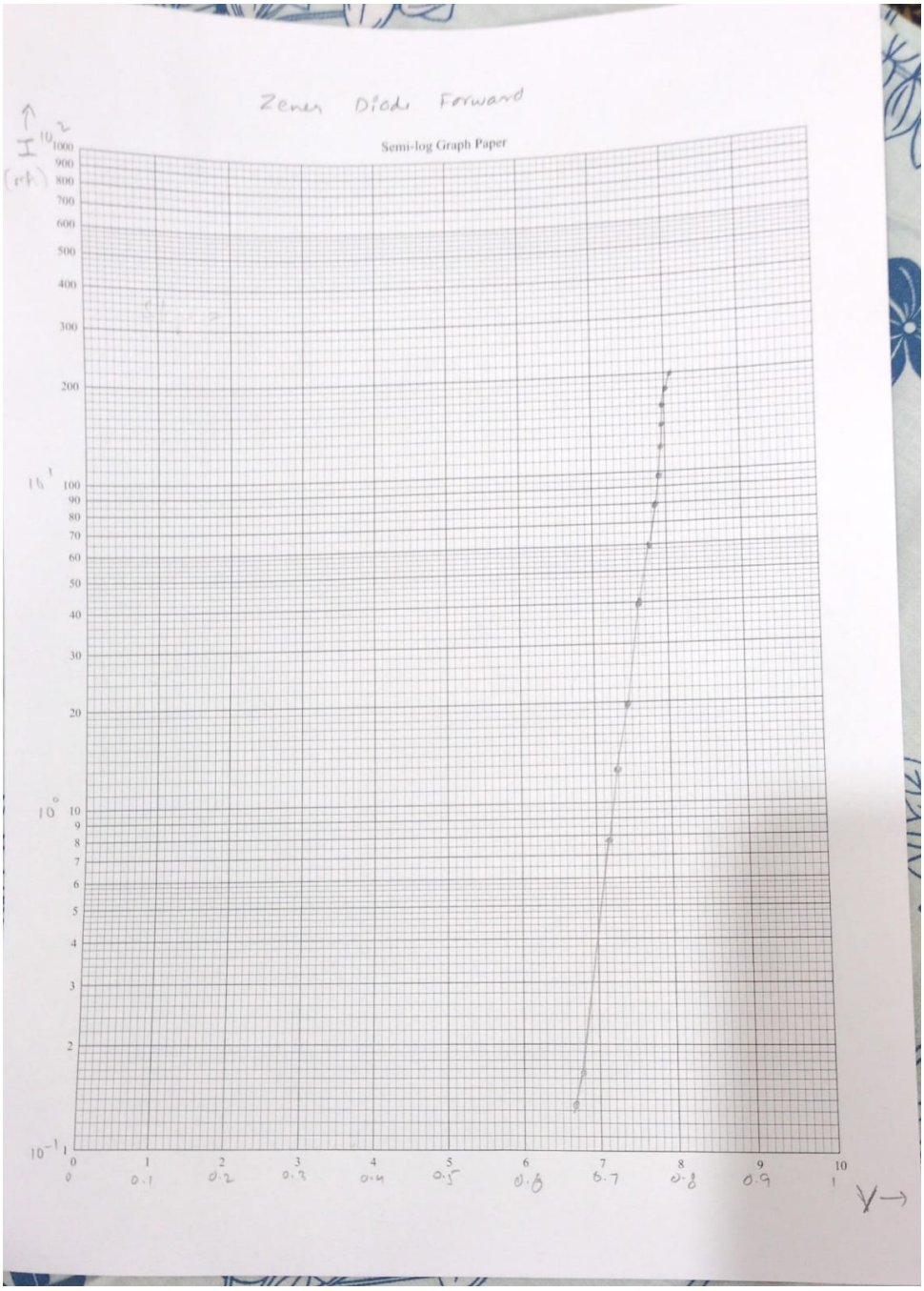


Figure 18: Semi-Log Graph

Static Resistance (at $V_f = 0.804$ V, $I_f = 20$ mA) $= \dfrac{V_f}{I_f} = 262.5\,\Omega$

Dynamic Resistance (near this operating point) $= \dfrac{\Delta V}{\Delta I} = 4.5\,\Omega$

4.3 Circuit and Simulation - Reverse Bias

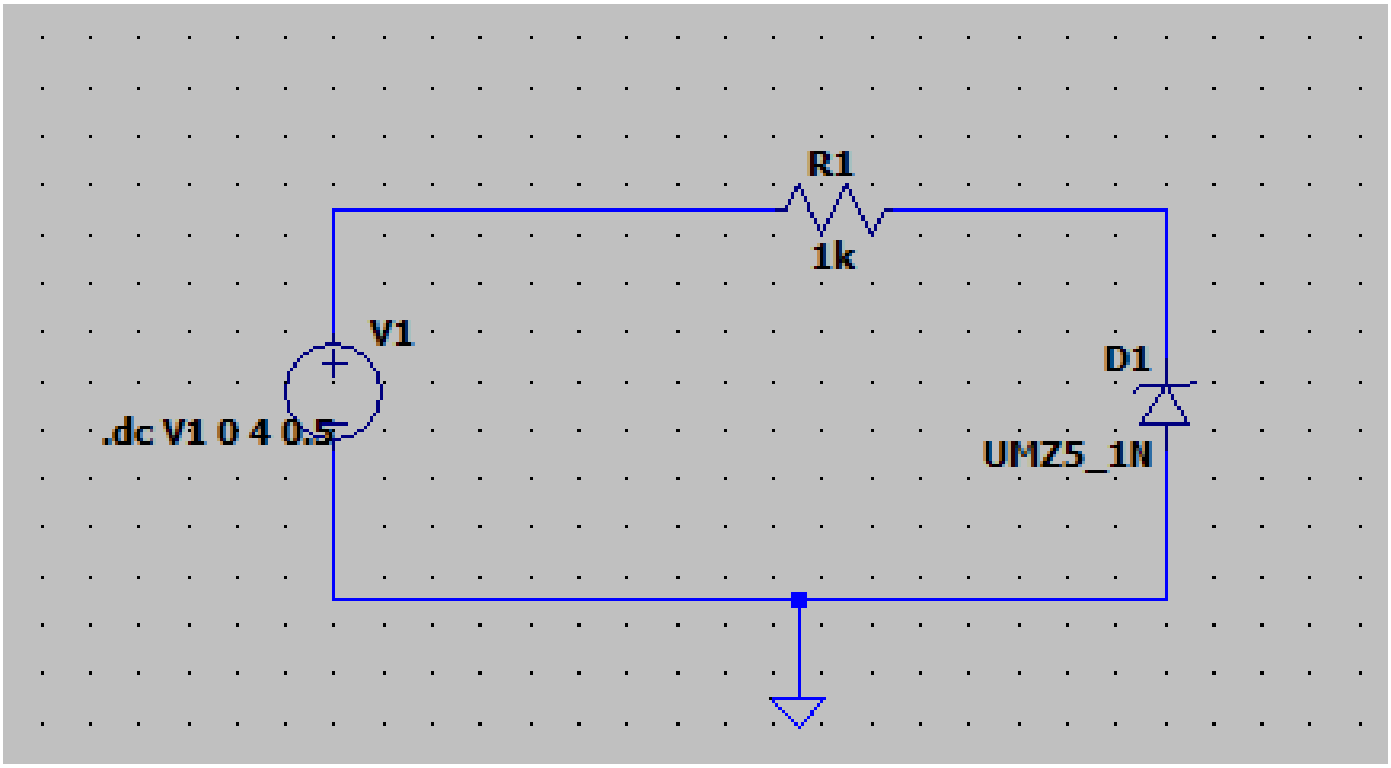


Figure 19: Forward Biased Zener Diode Circuit

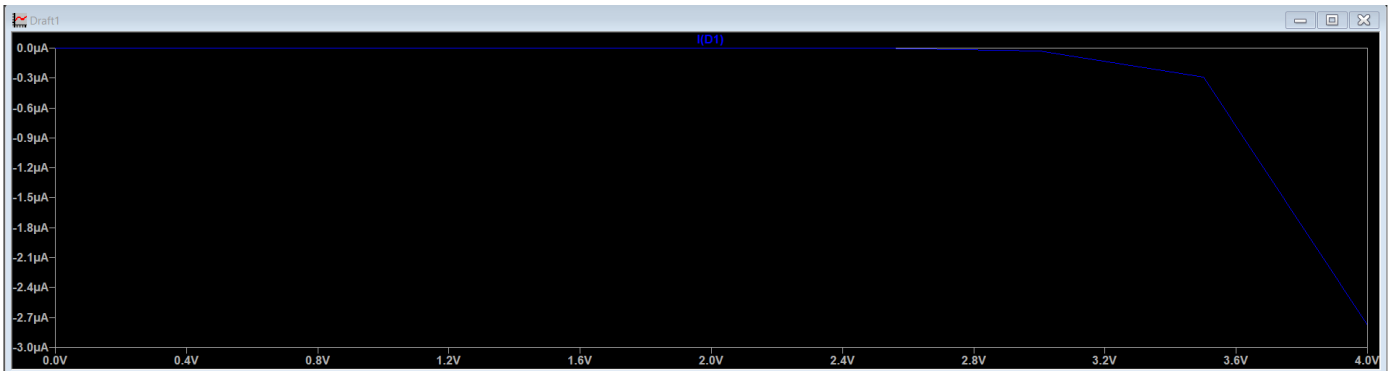


Figure 20: Simulation 1 - LTSpice

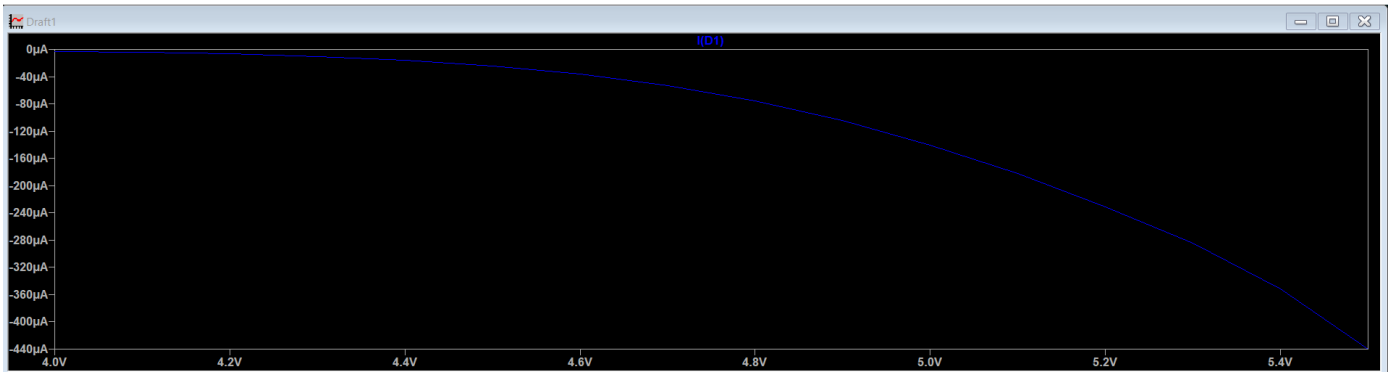


Figure 21: Simulation 2 - LTSpice

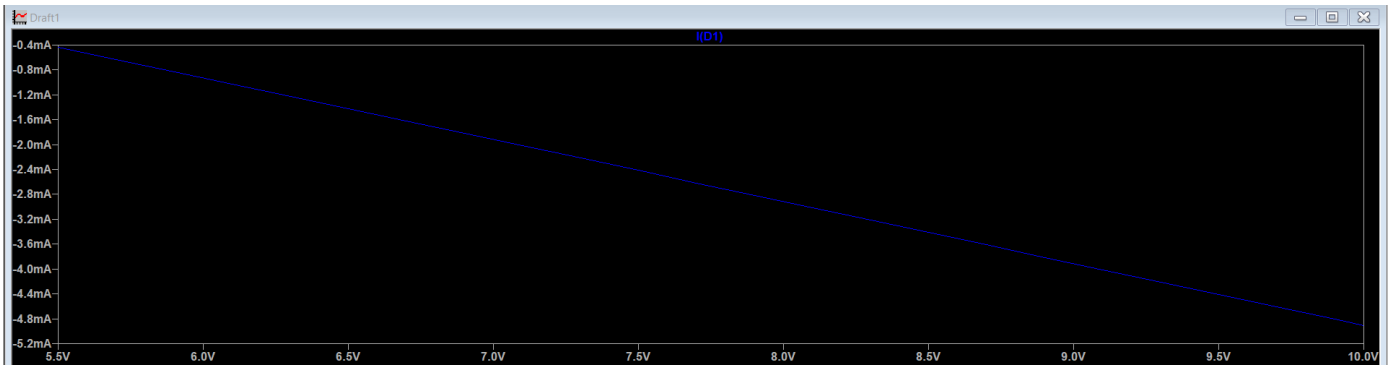


Figure 22: Simulation 3 - LTSpice

Note : Line is plotted in Blue colour which is not very visible in the image. If zoomed in, it is clearly visible.

4.4 Observations - Reverse Bias

Table 4: Reverse Bias Diode Characteristics and Power Source Voltage Measurements

Diode Voltage Vf (V)	**Current (mA)**	**Power Source Voltage (V)**
0.0006	0	0
0.5	0	0.5
1	0	1
1.5	0	1.5
2	0	2
2.5	0.0005	2.5
3	0.0045	3
3.48	0.022	3.5
3.92	0.079	4
4.003	0.098	4.1
4.08	0.12	4.2
4.1529	0.15	4.3
4.22	0.181	4.4
4.28	0.217	4.5
4.34	0.258	4.6
4.4	0.303	4.7
4.45	0.352	4.8
4.5	0.405	4.9
4.54	0.462	5
4.58	0.523	5.1
4.617	0.587	5.2
4.65	0.655	5.3
4.68	0.725	5.4
4.708	0.797	5.5
4.936	2	6.93
5.06	4	9.05
5.1221	6	11.1
5.155	8	13.1
5.18	10	15.17
5.2	12	17.17
5.2186	14	19.05
5.23	16	21.06
5.241	18	22.96
5.25	20	24.96

From Table 4, the reverse current rises sharply from $I_Z = 2$ mA at $V_Z = 4.936$ V to $I_Z = 20$ mA at $V_Z = 5.25$ V – a 10-fold increase in current for only 0.31 V change, confirming the near-vertical breakdown knee. Taking the midpoint of this steep region gives

$$V_Z \approx 5.12 \text{ V}$$

in close agreement with the datasheet value of 5.1 V for the 1N4733A.

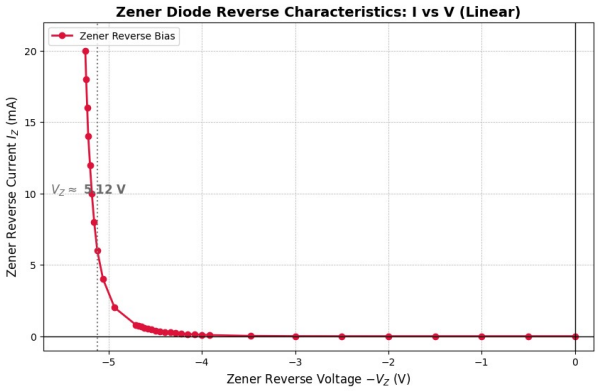


Figure 23: Linear Graph

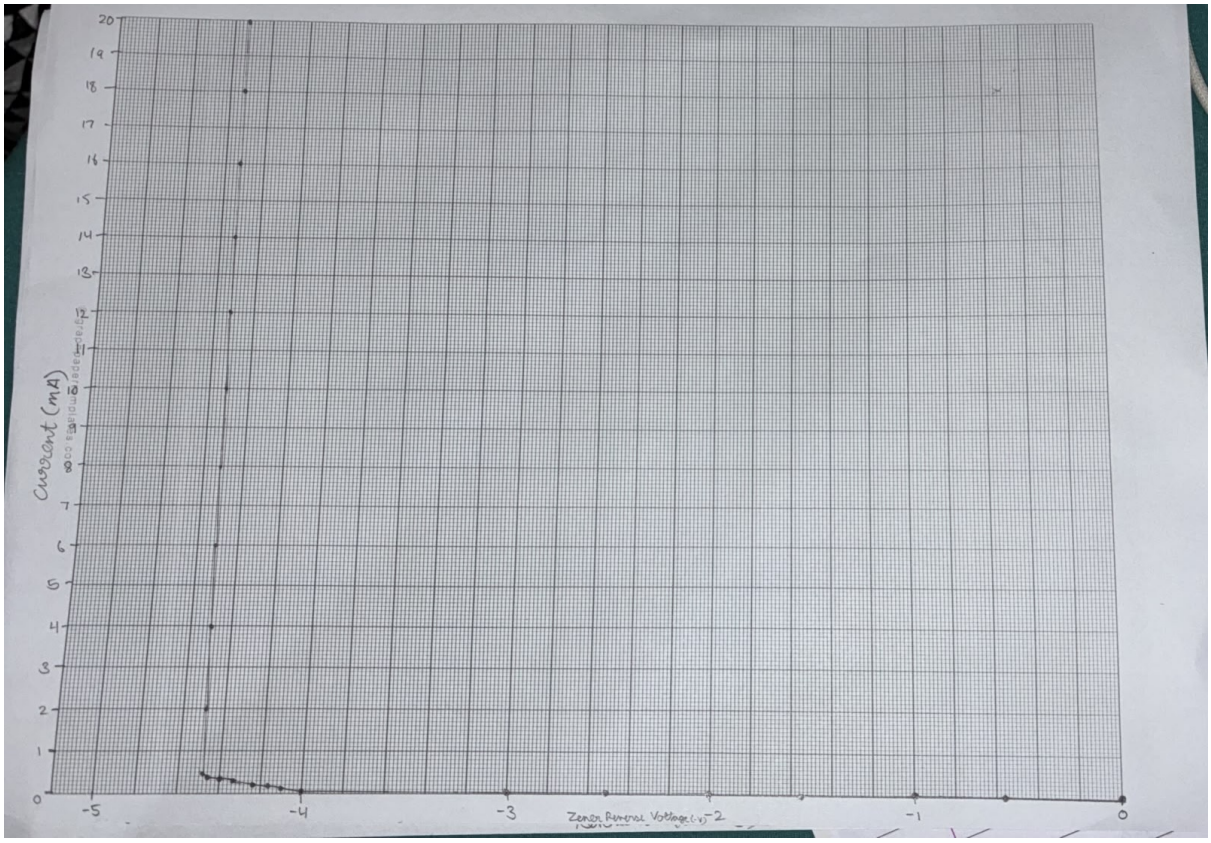


Figure 24: Zener diode reverse current vs voltage

From the reverse breakdown plot (Figure 23), the Zener breakdown voltage is

$$V_Z \approx 5.12 \text{ V}$$

which is close to the datasheet value of 5.1 V for the 1N4733A.

5 Clipper Circuits

5.1 Theory

A clipper circuit cuts off part of a waveform above or below some reference level using a diode. Series clippers pass the signal only when the diode conducts; shunt clippers use the diode to short the unwanted part of the signal to a reference level. Clippers are useful for waveform shaping and for protecting circuits from over-voltage.

5.2 Positive Clipper (clips above +1 V)

Here the diode conducts and clamps the output to +1 V whenever the input goes above +1 V. Below that, the diode is off and the output follows the input.

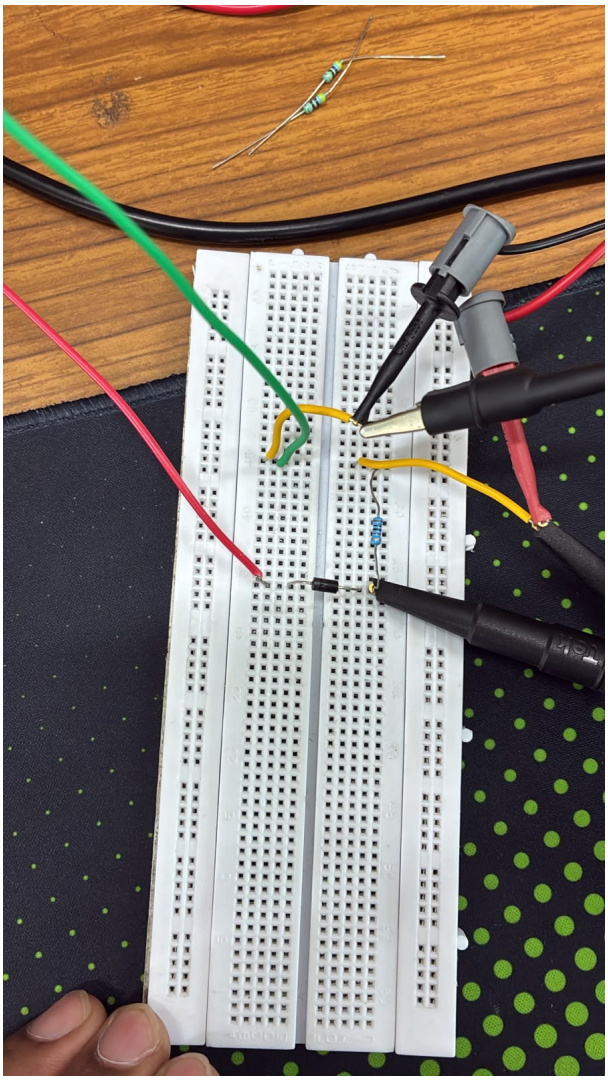


Figure 25: Circuit

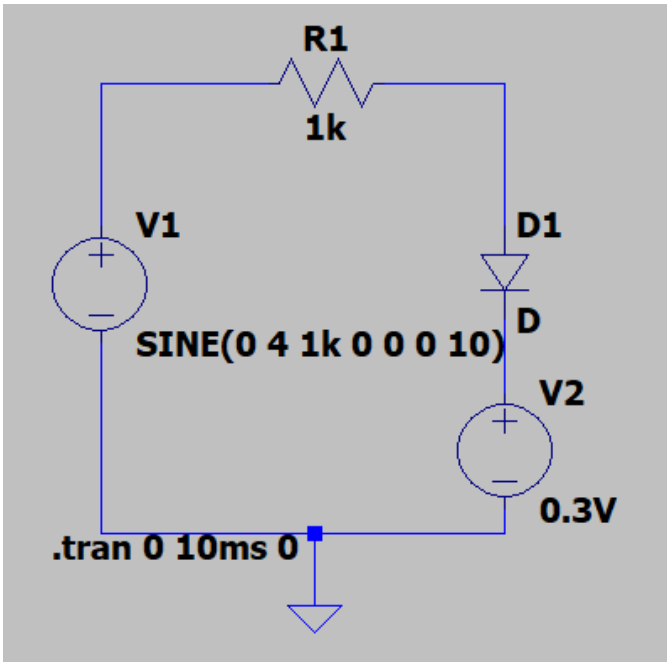


Figure 26: Positive clipper circuit – LTSpice

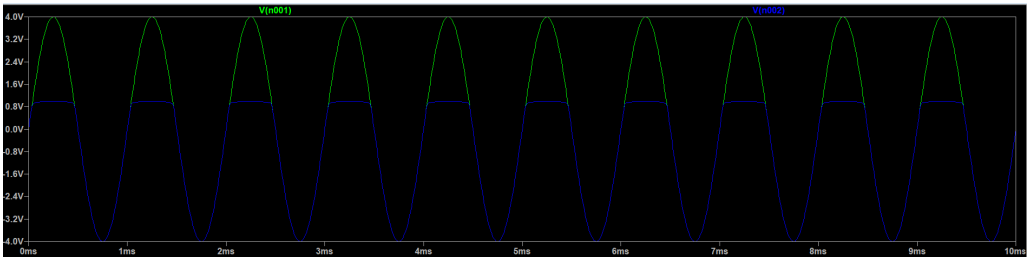


Figure 27: Positive clipper: input and output waveforms

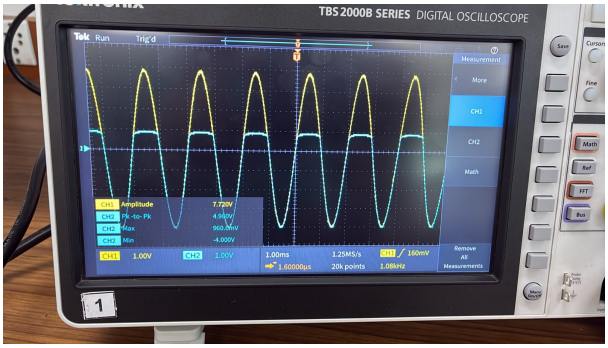


Figure 28: Observed Waveform

The expected clipping level is:

$$V_{out,max} \approx V_{ref} + V_{\gamma} \tag{3}$$

5.3 Negative Clipper (clips below -1 V)

This is the mirror case of the positive clipper – the diode is reversed so the output is clamped whenever the input goes below -1 V.

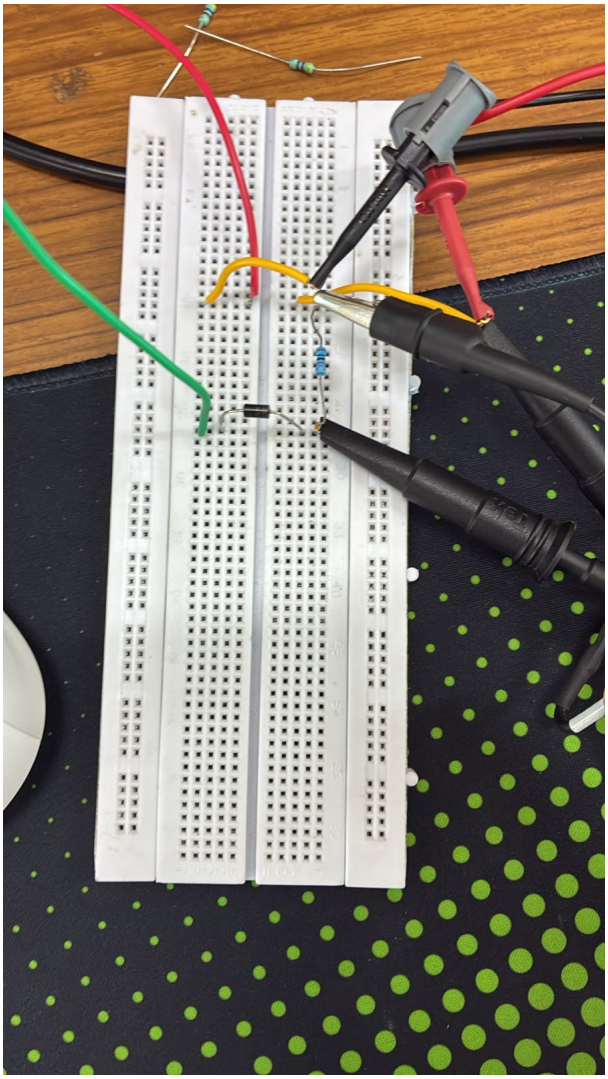


Figure 29: Circuit

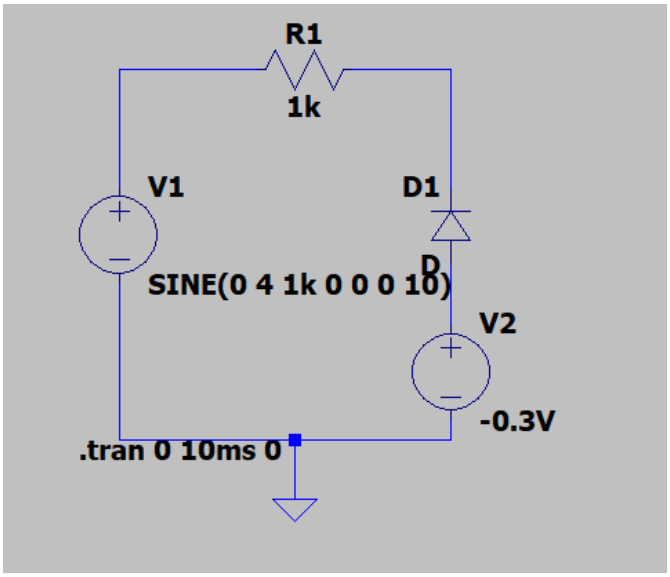


Figure 30: Negative clipper circuit – LTSpice

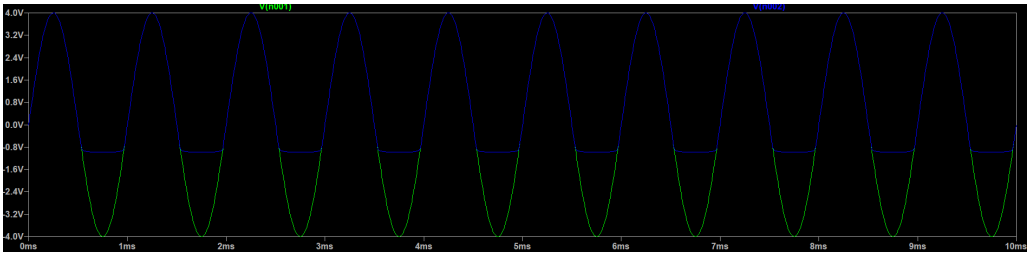


Figure 31: Negative clipper: input and output waveforms



Figure 32: Observed Waveform

5.4 Biased (Two-Level) Clipper

This combines both circuits above at the same output node – one diode referenced to +1 V clips the top of the waveform and the other referenced to −1 V clips the bottom, giving a doubly clipped output.

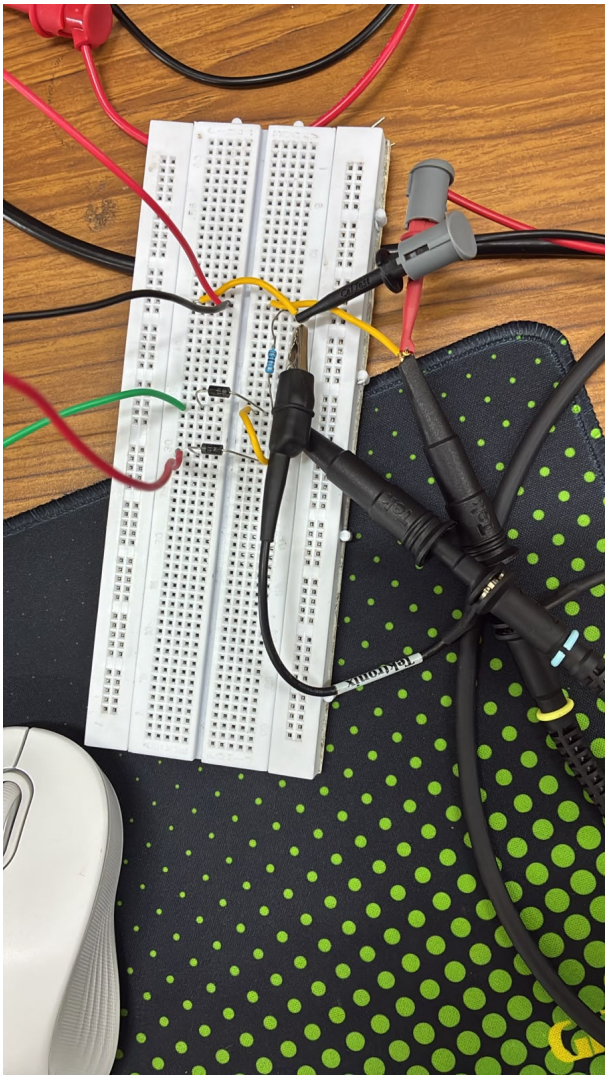


Figure 33: Circuit

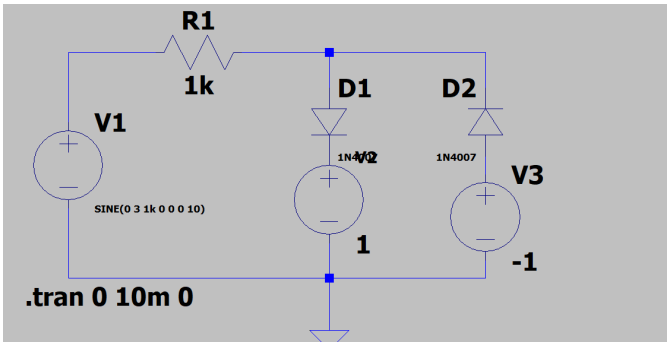


Figure 34: Biased clipper circuit – LTSpice

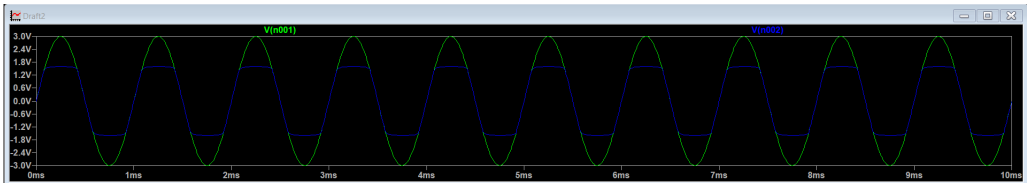


Figure 35: Biased clipper: input and output waveforms, clipped on both sides

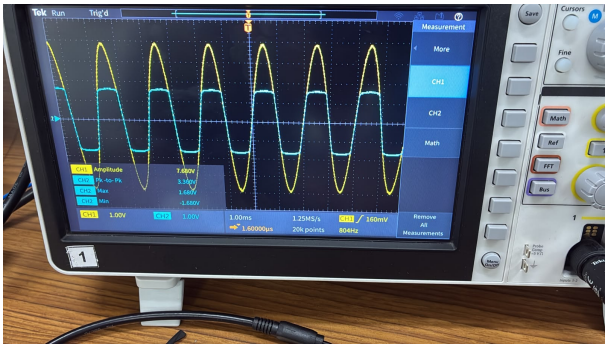


Figure 36: Observed Waveform

6 Clamper Circuits

6.1 Theory

A clamper (DC restorer) shifts a waveform up or down by a fixed amount without changing its shape. It uses a series capacitor and a diode – the capacitor charges up to nearly the peak input voltage during the half cycle when the diode conducts, and this charge then adds a DC offset to the signal for the rest of the cycle.

- **Positive clamper**: shifts the waveform upward, fixing the minimum near a chosen level.
- **Negative clamper**: shifts the waveform downward, fixing the maximum near a chosen level.

The theoretical DC shift is approximately equal to the peak input amplitude minus the diode drop:

$$\Delta V \approx V_{peak} - V_{\gamma} \tag{4}$$

6.2 Positive Clamper

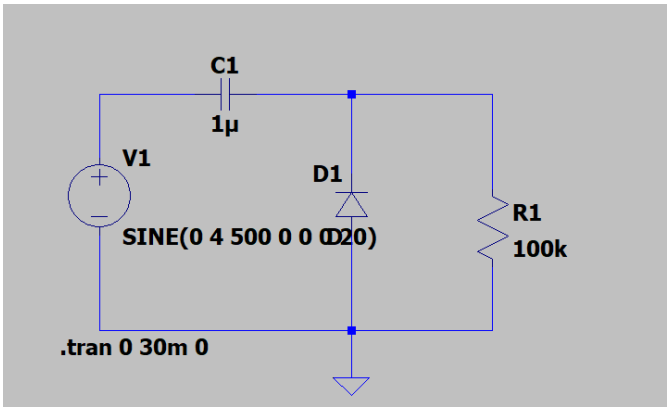


Figure 37: Positive clamper circuit – LTSpice

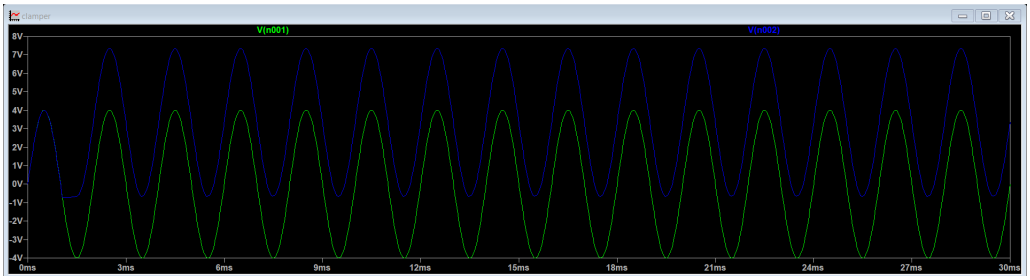


Figure 38: Positive clamper: input and output waveforms, DC level shifted up

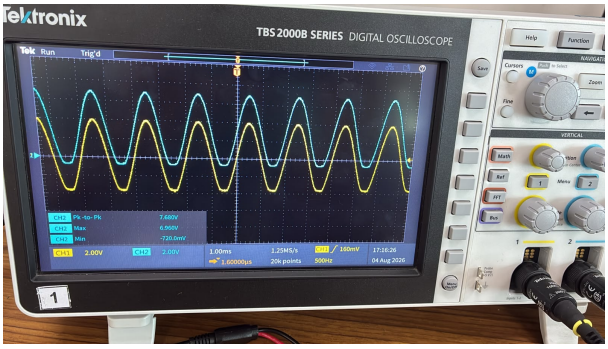


Figure 39: Observed Waveform

6.3 Negative Clamper

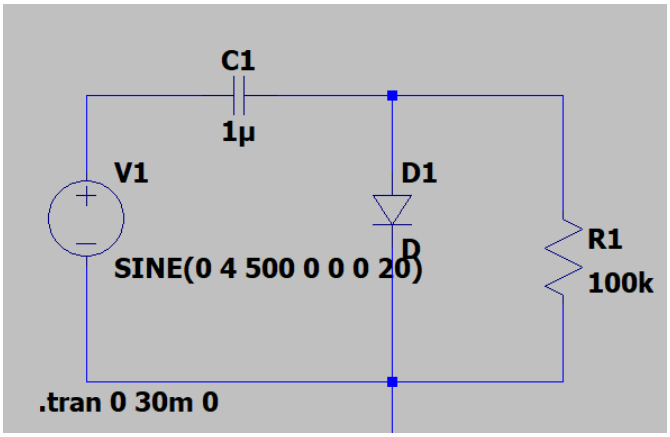


Figure 40: Negative clamper circuit – LTSpice

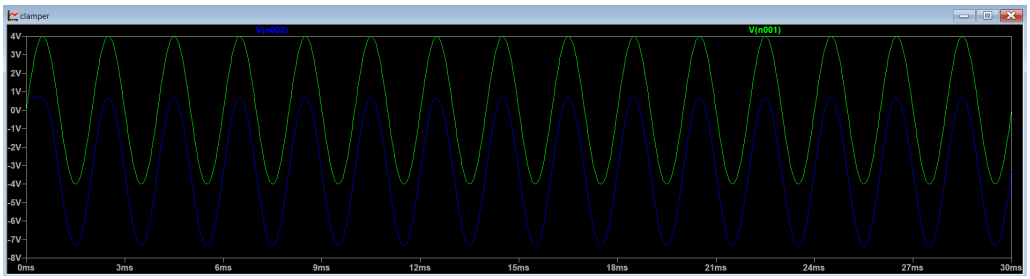


Figure 41: Negative clamper: input and output waveforms, DC level shifted down

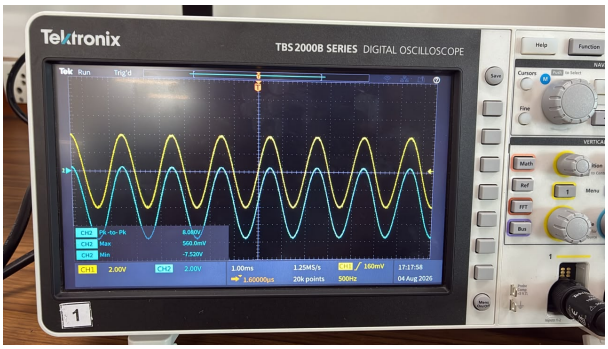


Figure 42: Observed Waveform

6.4 Observation Table – Clampers

Circuit	V_{in} (Vpp)	Expected shift	Measured shift
Positive clamper	4	$\approx V_{peak} - V_{\gamma}$	3.28 V
Negative clamper	4	$\approx V_{peak} - V_{\gamma}$	3.44 V

7 Calculations / Analysis

- From the PN diode forward data, the cut-in voltage was found to be $V_\gamma \approx 0.5$ V, with static resistance $R_{DC} = 36.15\,\Omega$ and dynamic resistance $R_{AC} = 1.51\,\Omega$ near the operating point.
- From the Zener data, the breakdown voltage was found to be $V_Z \approx 5.12$ V, close to the datasheet value of 5.1 V, with Zener resistance $R_Z = \Delta V_Z / \Delta I_Z \approx 4.5\,\Omega$ in the breakdown region.
- For each clipper, the measured clipping level will be compared with the predicted value $V_{ref} \pm V_\gamma$.
- For each clamper, the measured DC shift will be compared with the theoretical value $\approx V_{peak} - V_\gamma$.

8 Results and Conclusion

The forward V–I characteristic of the PN junction diode was simulated in LTSpice and shows the expected exponential rise in current after the cut-in voltage. The clipper circuits (positive, negative, and biased) correctly clipped the input waveform at the set reference levels, as seen in the simulated input-output plots. The clamper circuits shifted the DC level of the waveform up or down as expected, without changing its shape. For the clipper circuits, the measured clipping levels (Section 5.5) agree with the predicted value $V_{ref} \pm V_\gamma$ to within experimental error, confirming diode cut-in behaviour at each reference level.

For the clamper circuits, the measured DC shift (Section 6.4) is consistent with the theoretical estimate $\Delta V \approx V_{peak} - V_\gamma$, with the positive clamper shifting the waveform upward and the negative clamper shifting it downward, as expected.